A Prime-Residue Classification for a Central-Binomial Partial Sum

Primary author (AI): ChatGPT (OpenAI)

November 14, 2026

Abstract

For the OEIS sequence A066796, $a(n) = \sum_{k=1}^{n} \binom{2k}{k}$, we give a uniform congruence that proves and sharpens its stated divisibility interval for primes $p \equiv 1 \pmod 6$. This is an elementary explanation of a known OEIS observation, not a novelty claim.

OEIS assertion. A066796 states that for primes $p \equiv 1 \pmod 6$, every $a(n)$ with $(p-1)/2 \le n \le p-1$ is divisible by p.

Theorem 1. *Let $p > 3$ be prime and $(p-1)/2 \le n \le p-1$. Then*

$$a(n) \equiv \left(\frac{-3}{p}\right) - 1 = \begin{cases} 0 \pmod p, & p \equiv 1 \pmod 6, \\ -2 \pmod p, & p \equiv 5 \pmod 6. \end{cases}$$

Consequently, throughout this interval $p \mid a(n)$ if and only if $p \equiv 1 \pmod 6$.

Proof. For $0 \le k \le (p-1)/2$, reducing the factors in the polynomial binomial coefficient gives

$$\binom{2k}{k} = (-4)^k \binom{-1/2}{k} \equiv (-4)^k \binom{(p-1)/2}{k} \pmod p.$$

Here the first equality is an exact rational identity and the congruence is valid since $k!$ is invertible modulo p. For $(p+1)/2 \le k \le p-1$, the numerator $(2k)!$ contains a factor p, whereas $k!$ does not; hence $\binom{2k}{k} \equiv 0 \pmod p$. Therefore the value of $a(n)$ is constant modulo p on the asserted interval, and the binomial theorem yields

$$\begin{aligned} a(n) &\equiv \sum_{k=1}^{(p-1)/2} (-4)^k \binom{(p-1)/2}{k} \\ &= (1-4)^{(p-1)/2} - 1 = (-3)^{(p-1)/2} - 1 \pmod p. \end{aligned}$$

Euler's criterion identifies the power with the Legendre symbol $(-3/p)$. Quadratic reciprocity gives $(-3/p) = (p/3)$ for every odd prime $p \neq 3$, which is 1 if $p \equiv 1 \pmod 3$ and -1 if $p \equiv 2 \pmod 3$. The two odd-prime residue classes modulo 6 complete the proof. □

Scope. The result concerns congruences modulo p in the specified interval, not the stronger modulo-p^2 assertions also recorded in A066796. Its proof uses classical binomial congruences, Euler's criterion and quadratic reciprocity.

References

[1] OEIS Foundation, *A066796: Sum of central binomial coefficients*, https://oeis.org/A066796 (accessed November 14, 2026).

[2] G.-S. Mao and R. Tauraso, *Three pairs of congruences concerning sums of central binomial coefficients*, arXiv:2004.09155 (2020), https://arxiv.org/abs/2004.09155. (Related prior literature; no novelty claim.)

Authorship and verification. Primary author (AI): ChatGPT (OpenAI). This self-contained mathematical argument was prepared with automated arithmetic checks and source comparison. It has not received independent human peer review; no originality or priority is claimed.